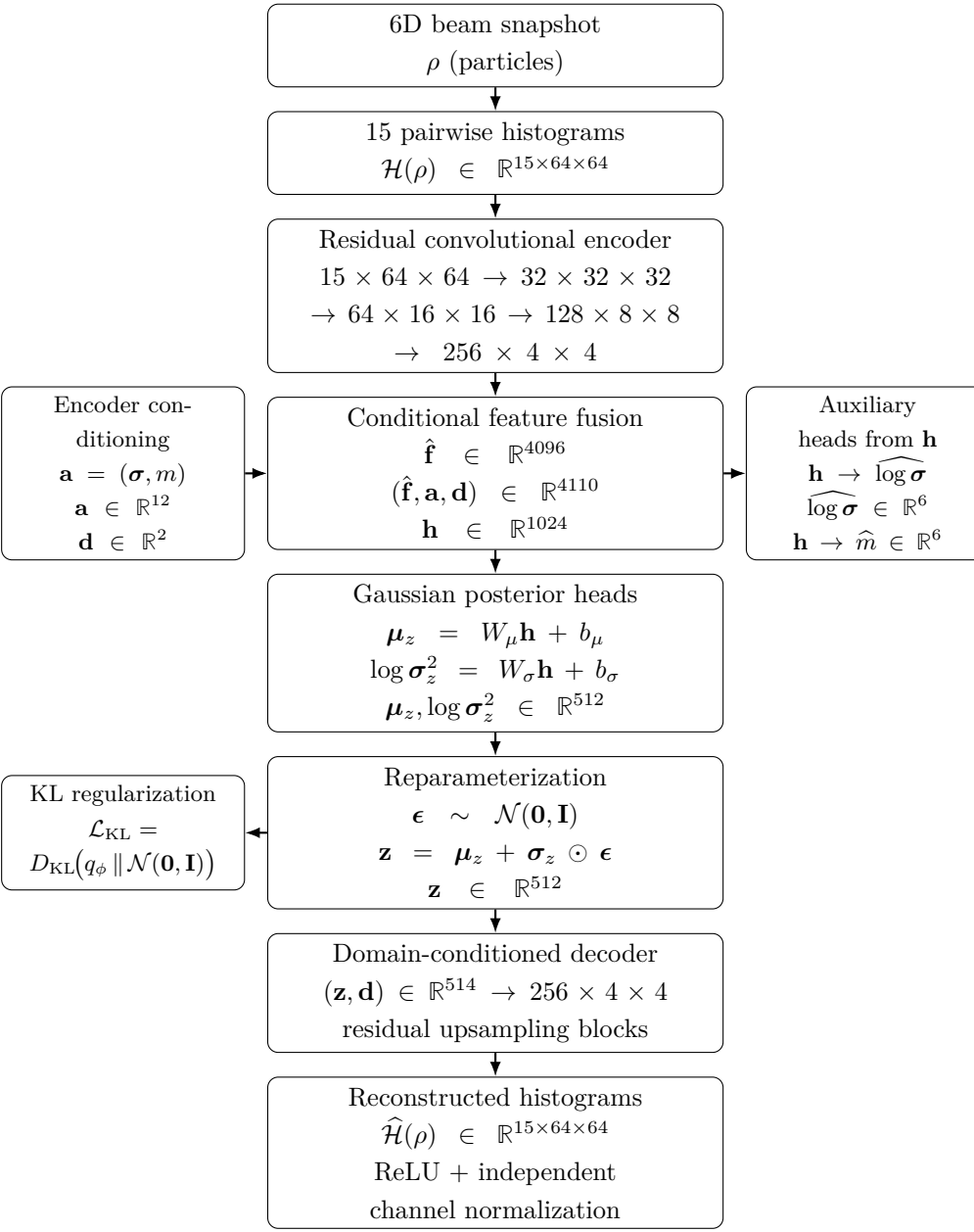

6D beam snapshot
ρ (particles)
15 pairwise histograms
$\mathcal{H}(\rho) \in \mathbb{R}^{15\times64\times64}$
Residual convolutional encoder
$15 \times 64 \times 64 \to 32 \times 32 \times 32$
$\to 64 \times 16 \times 16 \to 128 \times 8 \times 8$
$\to 256 \times 4 \times 4$
Encoder conditioning
$\mathbf{a} = (\boldsymbol{\sigma}, m)$
$\mathbf{a} \in \mathbb{R}^{12}$
$\mathbf{d} \in \mathbb{R}^2$
Conditional feature fusion
$\hat{\mathbf{f}} \in \mathbb{R}^{4096}$
$(\hat{\mathbf{f}}, \mathbf{a}, \mathbf{d}) \in \mathbb{R}^{4110}$
$\mathbf{h} \in \mathbb{R}^{1024}$
Auxiliary heads from $\mathbf{h}$
$\mathbf{h} \to \widehat{\log \boldsymbol{\sigma}}$
$\widehat{\log \boldsymbol{\sigma}} \in \mathbb{R}^6$
$\mathbf{h} \to \widehat{m} \in \mathbb{R}^6$
Gaussian posterior heads
$\boldsymbol{\mu}_z = W_\mu \mathbf{h} + b_\mu$
$\log \boldsymbol{\sigma}_z^2 = W_\sigma \mathbf{h} + b_\sigma$
$\boldsymbol{\mu}_z, \log \boldsymbol{\sigma}_z^2 \in \mathbb{R}^{512}$
KL regularization
$\mathcal{L}_{\mathrm{KL}} =$
$D_{\mathrm{KL}}\big(q_\phi \,\|\, \mathcal{N}(\mathbf{0}, \mathbf{I})\big)$
Reparameterization
$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
$\mathbf{z} = \boldsymbol{\mu}_z + \boldsymbol{\sigma}_z \odot \boldsymbol{\epsilon}$
$\mathbf{z} \in \mathbb{R}^{512}$
Domain-conditioned decoder
$(\mathbf{z}, \mathbf{d}) \in \mathbb{R}^{514} \to 256 \times 4 \times 4$
residual upsampling blocks
Reconstructed histograms
$\widehat{\mathcal{H}}(\rho) \in \mathbb{R}^{15\times64\times64}$
ReLU + independent
channel normalization